GCP Resource Tagging:
Enterprise Metadata Governance Platform

Prepared for: RB Global - Cloud Engineering | July 2026

1. Executive Summary

RB Global's GCP metadata governance solution is implemented as a **centralised, event-driven and serverless platform** for governing metadata across the Google Cloud estate. The platform supports both existing resources (**Brownfield**) and newly created resources (**Greenfield**) while keeping application ownership and governance values outside the runtime code.

Two dedicated governance projects provide strict environment isolation: **platform-metadata-dev** for engineering, integration and controlled validation, and **platform-metadata-prod** for production organisation-wide governance.

The design combines Cloud Asset Inventory, Cloud Audit Logs, Pub/Sub/Eventarc, Cloud Run, Cloud Tasks, Cloud Storage and BigQuery. The result is a pay-per-use control plane with no permanently running worker infrastructure, near-real-time Greenfield processing and auditable Brownfield remediation.

Core Platform Capabilities

Capability	Implementation
Brownfield Governance	Cloud Asset Inventory discovery -> compliance -> remediation planning -> Cloud Tasks -> worker -> resource adapter
Greenfield Governance	Cloud Audit Logs -> Pub/Sub/Eventarc -> Cloud Run -> classifier -> compliance -> resource adapter
Application Registry	YAML source of truth in Cloud Storage with schema validation and runtime caching
Compliance Engine	Compares actual metadata against registry-derived required metadata
Remediation	Service-specific adapters perform safe metadata updates using native GCP APIs
Audit & Reporting	BigQuery stores inventory, compliance, plans and execution evidence; dashboard consumes reporting APIs

2. Architecture and Processing Model

The same governance model is shared across Brownfield and Greenfield. Discovery/event ingestion differs, but both paths converge on registry resolution, capability validation, compliance evaluation, resource-specific remediation and audit evidence.

Processing Mode	End-to-End Flow
Brownfield	Target projects -> Cloud Asset Inventory -> Discovery -> Registry -> Compliance -> Planner -> Cloud Tasks -> Worker -> Adapter -> BigQuery
Greenfield	Resource creation -> Cloud Audit Logs -> Pub/Sub/Eventarc -> Cloud Run -> Classifier -> Capability -> Compliance -> Adapter -> BigQuery

Serverless Design Benefits

Benefit	Outcome
Cost Efficiency	Cloud Run and Cloud Tasks scale with actual work; there is no dedicated always-on remediation server.
Near Real-Time Compliance	Greenfield creation events can be evaluated shortly after the corresponding audit event is delivered.
Zero Idle Worker Infrastructure	Asynchronous workers are invoked only when remediation batches exist.
Resilience	Independent tasks, retries and failure isolation prevent one resource failure from stopping unrelated work.
Central Visibility	BigQuery and the dashboard provide one compliance and remediation view across governed projects.

3. Brownfield Governance

Brownfield processing governs existing resources across the selected project scope. Each execution creates a common **run_id**, discovers resources, evaluates only supported capabilities, plans required changes and submits remediation work asynchronously.

The current default remediation batch size is **500**. At that configuration, 1,000,000 planned resource actions represent approximately **2,000 Cloud Tasks batches** before retries or ancillary operations - not one million individual tasks.

Brownfield Control Points

Control	Behaviour
Discovery	Cloud Asset Inventory provides scalable inventory discovery across visible target projects.
Registry Resolution	Project binding resolves the application and required governance metadata.
Capability Gate	Unsupported or disabled asset types are excluded from mutation.
Compliance	Actual labels/metadata are compared with required values.
Planning	Only non-compliant resources requiring action become remediation plans.
Execution	Cloud Tasks delivers batches to the authenticated worker endpoint.
Evidence	Execution result, status and mode are persisted for audit/reporting.

Runtime Controls

Setting	Default	Purpose
REMEDIATION_BATCH_SIZE	500	Number of remediation actions grouped per task payload.
MAX_PARALLEL_WORKERS	10	Controls application-side parallel execution.
PRESERVE_EXISTING_LABELS	true	Protects unrelated existing labels during reconciliation.
DRY_RUN	false	Can be enabled for controlled onboarding and validation.
EXCLUDED_PROJECTS	Configured	Prevents selected projects from governance processing.
EXCLUDED_BUCKETS	Configured	Protects registry/platform buckets from unintended remediation.

4. Greenfield Governance

Greenfield governance is event-driven. Resource creation activity is captured from Cloud Audit Logs and routed through the organisation-level event path into the governance service. The event is normalised by a classifier before the shared capability, compliance and remediation layers are invoked.

A client/adaptor existing in the repository does not by itself mean that a resource is Greenfield-supported. A resource is enabled only after the exact creation event, routing, classifier, capability entry, adapter behaviour, IAM and retry/idempotency characteristics have been validated.

Greenfield Gate	Required Validation
Audit Event	Exact serviceName and methodName for creation are confirmed.
Organisation Routing	Audit event from governed workload projects reaches the central transport path.
Classifier	Event is mapped to the correct canonical resource and asset type.
Capability	Asset type is explicitly enabled in supported resource configuration.
Adapter	Read and metadata mutation behaviour is tested for the target service.
IAM	Runtime service account has only the permissions required by enabled resources.
Idempotency	Duplicate event delivery is safe and does not create destructive changes.

5. Registry, Data and Reporting

The **Application Registry** is the metadata source of truth. Application ownership and governance values are stored as YAML in Cloud Storage and validated against the registry schema before promotion. Runtime caching reduces repeated object reads.

Typical registry attributes include product, team, owner, budget owner, organisation, department and cost centre, with GCP bindings defining project, region, environment and business criticality.

BigQuery Governance Data

Object	Purpose
resource_snapshot	Discovered inventory and resource metadata snapshot.
compliance_snapshot	Point-in-time compliance evaluation evidence.
remediation_plan	Planned changes generated for a governance run.
remediation_execution	Execution status, evidence and BROWNFIELD/GREENFIELD execution mode.

Dashboard and Reporting

View	Purpose
Executive Summary	Estate size, supported resources, compliance and overall governance position.
Brownfield Summary	Planned, completed, remaining and failed remediation.
Greenfield Summary	Events, successful/failed processing and recent activity.
Project View	Compliance breakdown by governed project.
Resource Type View	Compliance by supported GCP resource family.
Recent Runs	Operational execution history.
Top Non-Compliant Resources	Prioritised resources with missing or incorrect governance metadata.

6. Resource Capability Model

Runtime enablement is centrally controlled by the supported-resource configuration. This separation is important because the repository can contain clients or adapters that are under development, disabled, or unsuitable for labels.

The implementation contains integration code for multiple GCP service families, including Compute Engine, Cloud Storage, BigQuery, Cloud SQL, GKE, Pub/Sub, Artifact Registry, Cloud Run, App Engine, Redis, Cloud KMS, Secret Manager, Monitoring, Eventarc and other platform services. **Production support must be claimed only for resource types explicitly enabled and validated in the active capability configuration.**

Capability Governance

Rule	Requirement
Code Presence != Support	An adapter/client in the repository is not sufficient evidence of production support.
Explicit Enablement	The asset type must be present in the active label/tag capability set.
Test Before Promotion	Brownfield and, where applicable, Greenfield behaviour must be validated in platform-metadata-dev.
IAM Follows Capability	Permissions are granted for enabled resource types rather than every candidate integration.
Tags Separate from Labels	Resource Manager Tag capability is governed independently from service label capability.

7. Dedicated Project and Environment Model

Area	platform-metadata-dev	platform-metadata-prod
Purpose	Development, integration, adapter/event testing and controlled scale validation	Production organisation-wide metadata governance
Runtime Identity	Dedicated DEV service account	Dedicated PROD service account
Registry	DEV registry/configuration	Production-approved registry/configuration

Area	platform-metadata-dev	platform-metadata-prod
BigQuery	DEV governance dataset	Production governance/audit dataset
Event/Queue Infrastructure	DEV-specific	PROD-specific
Access Boundary	Must not unintentionally remediate PROD workloads	Independent production trust boundary

8. IAM and Security Model

The governance runtime uses a dedicated service account and follows least privilege. The final client custom role should be generated from the **resource capabilities actually enabled for production**, not simply from every API client present in the repository.

Permission Categories

Category	Required Access
Organisation / Project Discovery	Read access required to enumerate authorised target projects and resolve project metadata.
Cloud Asset Inventory	Read/search access required for Brownfield resource discovery.
Registry	Read access to the approved Cloud Storage registry objects.
BigQuery	Required dataset/table read and write access for governance evidence and reporting.
Cloud Tasks	Permission to enqueue remediation batches and authenticated invocation for the worker path.
Resource Metadata	Service-specific read/update permissions only for enabled adapters.
Resource Manager Tags	Tag permissions only when Tag governance is explicitly enabled.
Dashboard	Restricted to authorised organisation users; operational worker endpoints remain service-to-service.

9. Approach Comparison

Criteria	Earlier Native-Only Business Case	Current Enterprise Platform
Runtime	**No application runtime**	**Modular Cloud Run governance engine**
Brownfield Detection	**CAI export + BigQuery SQL**	**CAI discovery + live metadata enrichment + compliance engine**
Brownfield Remediation	**Terraform/gcloud operational workflow**	**Planner + Cloud Tasks + worker + service-specific adapters**
Greenfield	**Terraform + Organisation Policy prevention**	**Near-real-time audit-event-driven detection and remediation**
Registry	**Application Registry used as value source**	**Validated YAML source of truth loaded dynamically from Cloud Storage**
Reporting	**BigQuery + Looker Studio concept**	**BigQuery-backed reporting APIs and integrated dashboard**
Scale Model	Operator/script based	Asynchronous serverless batch processing

10. Recommendation

Proceed with the **Enterprise Metadata Governance Platform** using **platform-metadata-dev** as the mandatory validation environment and **platform-metadata-prod** as the production control plane.

Production onboarding should be capability-by-capability. Each resource family must have verified discovery, metadata behaviour, least-privilege IAM, retry/idempotency and - for Greenfield - exact creation-event routing before it is enabled.

Production Readiness Gate	Required Outcome
Registry	Schema validation and unique project bindings pass before promotion.
Capability	Only tested resource types are enabled.
IAM	Least-privilege permissions match enabled capabilities.
Brownfield	Batching, retry, failure isolation and representative scale are validated.
Greenfield	Exact creation events and end-to-end organisation routing are validated.
Security	Dashboard access and internal service endpoints are authenticated and restricted.
Operations	Logging, alerting, budgets, rollback and runbook are approved.

> **Important - Capability Control:** Resource support is determined by the active capability configuration. The presence of an adapter, client or IAM permission must never be treated as proof that a resource is production-enabled.

Architecture position: Event-driven | Serverless | Registry-driven | Auditable | Cost-efficient | DEV/PROD isolated